Mike Hall

Staff Observability & Resilience Architect

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Professional Summary

Observability and Resilience Architect specializing in illuminating dark telemetry, enterprise OpenTelemetry rollouts, and deep incident root cause triage. Eliminates cross-service blind spots by engineering end-to-end distributed trace propagation across legacy microservices and modern cloud infrastructure.

Core Skills

OpenTelemetry Working Group Leadership & Governance, Distributed Trace Context Propagation (W3C Trace Context), OTel Collector Pipeline Engineering (Receivers, Processors, Exporters), Dark Telemetry Illumination & Latency Bottleneck Detection, Edge PII Scrubbing, Masking & Sensitive Data Governance, Production Incident Triage for 100+ Person Bridge Calls, APM & Backend Integration (OpenObserve, Datadog, ELK, Prometheus, PgHero), High-Concurrency Distributed Systems Reliability

Experience

Staff Software Engineer

just3ws (Independent) | Chicago, IL | February 2026 - Present

Instrumented Rails 8, Solid Queue, PostgreSQL 18 (pgvector), and TypeScript/Node.js WebSocket platforms with OpenTelemetry trace propagation, structured LTSV wide logging, PgHero query telemetry, and property-based verification gates.

- **Autonomous Media Pipeline & Local AI RAG Microservice:** Architected a Rails 8, Solid Queue, and PostgreSQL 18 (pgvector) media ingestion and publishing platform with on-device Metal GPU Whisper transcription, hybrid lexical and vector semantic search via Reciprocal Rank Fusion (RRF), local LLaMA 3.1 RAG synthesis via RubyLLM, OpenTelemetry distributed tracing, Ahoy analytics, and PgHero query tuning across 270+ processed video masters.
- **Forensic Audio DSP Restoration & Sovereign Verification (dehum):** Engineered a deterministic 5-stage FFmpeg audio restoration and clock-drift synchronization pipeline (atempo, adeclick, highpass, afftdn, and 8-harmonic 60 Hz notch cascade) eliminating electromagnetic interference and corona discharge crackle recorded under 345,000-volt transmission lines while preserving lossless video streams.
- **Technical Conversation Archive & Knowledge Graph:** Built and curated a contract-validated oral history platform preserving 218 technical interviews with Software Craftsmanship, Ruby, and JVM practitioners, enforcing build-time schema verification, speaker diarization, and semantic knowledge graph cross-linking across 500+ indexed pages.
- **Deterministic State Engine & Verification Platform (Phalanx Duel):** Engineered a server-authoritative real-time tactical game platform in TypeScript, Node.js, PostgreSQL, and WebSockets featuring a pure deterministic rules engine, append-only replayable action ledger, OpenTelemetry trace propagation, and property-based CI verification gates across the complete state lifecycle.
- **Multi-Source Ingestion & Prompt-Injection Defense (WWWorkRemote):** Designed a Rails 8 and pgvector semantic matching platform with a Manifest V3 Chrome extension across 16 ATS adapters and a 4-stage prompt-injection guardrails pipeline screening untrusted scraped text before local LLM evaluation.

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Founded and led the OpenTelemetry Working Group, establishing distributed observability across application and integration boundaries and connecting system signals to incident response and safer change.

- **Originations IC Delivery & Instant Disbursement:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping an instant loan disbursement pipeline (funding to debit cards in minutes). Appointed Software Architect for the Acquisition Lane, later converting to Associate Director, Staff Engineer.
- **Acquisition Workflow Enablement & Architecture Discovery:** Founded the Acquisition Lane enablement team; mapped the loan acquisition and origination pipeline end-to-end from visit to booked loan (desktop browser to mainframe and back again at each step) and the Affiliate API at the same depth, identifying seven distinct lending funnels with a tracking initiative; architected an automated 5-phase PII remediation deletion engine across 30+ tables.
- **Enterprise Resilience & DynamoDB Session Remediation:** Diagnosed and eliminated a critical session-cookie overflow defect that caused 4% silent traffic loss of digital loan applications during e-signing; architected zero-downtime blue/green migration to DynamoDB session store with zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends; partnered directly with Cybersecurity, EMC, SRE, and Incident Command to align monitoring with end-to-end distributed tracing.

- **Community Enablement & SRE Handoff:** Founded weekly Geekfest@OMF and scaled the enterprise OpenTelemetry Working Group to 40+ cross-lane engineers; mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable community of practice.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons building conversational agents with Rasa and a schema inference prototype; pioneered local LLM orchestration and privacy-conscious AI developer tooling at Geekfest@OMF.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Standardized asynchronous Sidekiq error observability, queue latency metrics, and automated deployment alerts.

- **Search Infrastructure Modernization:** Eliminated external cluster failure modes and reduced operating footprint by migrating legacy Sphinx to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Rails Upgrades:** Led Ruby and Rails framework upgrades across core applications, sequencing database migrations to ensure zero downtime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Made previously opaque PHP backend behavior observable through structured query patterns and test coverage, improving change safety and developer feedback loops.

- **Legacy Query Modernization:** Replaced ad-hoc global data access with structured, cacheable query patterns in the PHP backend, enabling unit testability and sub-second query execution.
- **Frontend Verification Gates:** Hardened Ember.js automated test suites across core CRM contact workflows, creating safety gates that enabled non-breaking legacy PHP refactoring.

Senior Software Developer

BenchPrep | Chicago, IL | March 2017 - February 2018

Optimized high-concurrency PostgreSQL scoring queries and implemented cryptographic JWT payload verification to eliminate calculation drift.

- **Assessment Query Optimization:** Optimized complex PostgreSQL calculations across high-stakes exam scoring engines, eliminating calculation drift and reducing query latency.
- **JWT Response Verification:** Implemented JWT-based cryptographic payload signing and once-only processing, securing exam submission pipelines against client-side tampering.
- **Secure Embedded Client APIs:** Designed secure cross-origin API boundaries for embedded enterprise assessment clients, establishing explicit authentication and trust contracts.

Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)

Groupon | Chicago, IL | July 2011 - May 2013

Engineered fraud detection queries in Vertica and conducted exploratory anomaly detection spikes in Hadoop and Clojure.

- **Centralized Taxonomy Service:** Implemented merchant taxonomy service in Java and MySQL, eliminating categorization drift across distributed product teams during hypergrowth.
- **Analytical Fraud Detection:** Built analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure for transaction anomaly scoring.
- **Merchant Analytics Pipelines:** Built high-throughput merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- **Global Engineering Enablement:** Redesigned technical onboarding curricula, standardizing engineering practices and shortening time-to-first-commit for 100+ global engineering hires.

Earlier Career Experience

- **Principal Consultant** | **Tandem** | Chicago, IL | August 2018 - August 2018: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.
- **Senior Software Developer** | **ReachLocal** | Remote | March 2015 - November 2016: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.

- **CTO** | **KloboMedia** | Remote | March 2014 - September 2016: Owned end-to-end platform architecture, leading high-throughput data integration for an early-stage social media startup. The product, TheSocReport, ended when the upstream data taps closed: Twitter's firehose shutdown and the closing of Facebook's and Instagram's user-intelligence APIs removed the inputs the analytics depended on.
- **Open-Source Transition Lead** | **Coderwall** | Remote | January 2014 - December 2014: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Software Engineer** | **Obtiva** | Chicago, IL | August 2009 - July 2011: Delivered backend systems and services as a consultant across multiple clients, including Sears, Leapfrog Online, and Groupon, supporting early-stage scaling and data-driven operations in high-growth environments.
- **Assistant Vice President, Application Developer** | **JPMorgan Chase & Co.** | Chicago, IL | April 2008 - October 2008: Owned cross-system integration stability, leading defect resolution in regulated financial workflows.
- **Senior .NET Developer** | **Brightstar Corporation** | Libertyville, IL | March 2007 - April 2008: Owned enterprise supply-chain integration, leading cross-system data unification for a global mobile device distributor.
- **Senior .NET Developer** | **TicketsNow** | Crystal Lake, IL | November 2005 - March 2007: Owned real-time inventory systems, leading transactional integrity and iterative delivery for revenue-critical operations.

Technical Leadership

- Established enterprise-wide OpenTelemetry standards and transitioned ownership after adoption stabilized across teams
- Realigned Acquisition engineering responsibilities around business ownership, then founded and technically led an enablement team to sustain platform risk and remediation work
- Built Communities of Practice to prevent architectural drift and align cross-team decisions
- Built and ran global engineering onboarding and training at Groupon during hypergrowth, improving ramp time and retention
- Founded Software Craftsmanship McHenry County (2009) with Ryan Gerry and Jim Suchy, building a community that still meets with 820+ members
- Founded WHOIS Tech Community (UGtastic), interviewing industry leaders and shaping software craftsmanship discourse